

[Sophia Exclusive] Conceptual Rigor & Careless Mistake Elimination (Set 1)

Name:

Sophia

Target:

 **Focus: Square Numbers / Fraction Expansion / Draft Alignment**

Time: 30

Mins

Total: 100

Pts

Score:

Teacher's Key Reminders (Read before starting):

- ① **Square Numbers:** Must be the product of two *identical integer twin factors* (e.g. $3 \times 3 = 9$). A rectangle $2 \times 3 = 6$ is NOT a square number!
- ② **Area vs Side Length:** The side length of a square with area A is $\sqrt{A}$. Never divide the area by 2!
- ③ **Fractions:** When multiplying the denominator to find a common denominator, you **MUST** multiply the numerator by the exact same number!
- ④ **Two-Column Work:** Always show vertical calculation steps in the right scratch column to prevent impulsive arithmetic slips like $18 + 5 = 90$.

Section 1: Conceptual Precision & True Meaning (5 pts each, 25 pts total)

Score: _____

1. Circle all the **Square Numbers (Perfect Squares)** below, and write their base square representations (e.g. $16 = 4^2$):

1

4

6

9

12

16

18

24

25

36

48

49

64

72

81

Your recorded square numbers with their base representations:

Example: $16 = 4^2$;

2. [True or False Reasoning] A student argues: "*Because $6 = 2 \times 3$, it is the product of two integers, so 6 must be a square number.*" Is this statement correct? Explain your reasoning in 1-2 sentences:

Judgment: _____ (True / False)

Reason:

3. [Geometry Area Reversal] A square has an area of 36 cm^2 .

(a) What is the side length of this square? (Show your reasoning; DO NOT divide by 2)

(b) What is the perimeter of this square?

(a) Side length = _____, because (_____)

(b) Perimeter = _____

4. A rectangle has length 8 cm and width 2 cm.

(a) Find its area. (b) Can this rectangle be reshaped into a square with the exact same area? If yes, what is the side length of the square?

(a) Rectangle area = _____

(b) Square side length = _____

5. [AMC8 Focus · Prime Factors and Powers] The integer 180 is multiplied by a positive integer k to give a perfect square. Find the smallest possible value of k .

Hint: Prime factorize $180 = 2^a \times 3^b \times 5^c$. In a perfect square, all prime factor exponents must be EVEN!

Prime factorization: $180 =$ _____

To make all exponents even, the missing factor $k =$ _____

Section 2: Equivalent Fractions & Common Denominators (5 pts each, 25 pts total) **Score:** _____

6. [Dual-Sync Expansion] According to the fundamental property of fractions, the numerator and denominator MUST be multiplied by the same non-zero number. Fill in the blanks:

(a) $\frac{3}{4} = \frac{3 \times \underline{\hspace{1cm}}}{4 \times 5} = \frac{\underline{\hspace{1cm}}}{20}$ (b) $\frac{2}{5} = \frac{2 \times \underline{\hspace{1cm}}}{5 \times 4} = \frac{\underline{\hspace{1cm}}}{20}$

(c) Therefore: $\frac{3}{4} + \frac{2}{5} = \frac{\underline{\hspace{1cm}}}{20} + \frac{\underline{\hspace{1cm}}}{20} = \frac{\underline{\hspace{1cm}}}{20} = \underline{\hspace{1cm}}$

7. [Error Diagnosis & Correction] A student solved the following subtraction:

Mistake: $\frac{5}{6} - \frac{1}{4} = \frac{5-1}{12} = \frac{4}{12} = \frac{1}{3}$

Point out the conceptual error and show the correct step-by-step calculation:

Error: _____

Correct step-by-step solution:

$$\frac{5}{6} - \frac{1}{4} =$$

8. Evaluate each expression below (Show full common denominator steps; NO skipping):

FORMAL SOLUTION (LEFT COLUMN)

(a) $\frac{5}{8} + \frac{1}{6} =$

SCRATCH WORK (RIGHT COLUMN)

Find LCM[8, 6]:

$$8 = 2^3$$

$$6 = 2 \times 3$$

$$\text{LCM}[8, 6] =$$

FORMAL SOLUTION (LEFT COLUMN)

(b) $2\frac{1}{3} - 1\frac{3}{4} =$

SCRATCH WORK (RIGHT COLUMN)

Mixed to improper:

$$2\frac{1}{3} = \frac{\quad}{3}$$

$$1\frac{3}{4} = \frac{\quad}{4}$$

9. [Smart Fraction Comparison] Compare $\frac{11}{13}$ and $\frac{15}{17}$ (Hint: DO NOT cross-multiply large numbers; use the "distance to 1" benchmark method).

$$1 - \frac{11}{13} = \underline{\hspace{2cm}}, \quad 1 - \frac{15}{17} = \underline{\hspace{2cm}}$$

Because ($\hspace{2cm}$), therefore $\frac{11}{13} \text{ — } \frac{15}{17}$

Section 3: Two-Column Scratch Work & Anti-Slip Practice (10 pts each, 20 pts total)

Score: _____

⚠ Rule: The left column must contain the structured algebraic steps. The right column **must** retain vertical scratch calculations. Skipping scratch work deducts 4 points!

10. Evaluate: $18 + 5 \times (14 - 8) \div 3$

STEP-BY-STEP SOLUTION

=

=

SCRATCH VERIFICATION COLUMN

- ① Parentheses: $14 - 8 =$
② Multiplication: $5 \times (\quad) =$
③ Division: $(\quad) \div 3 =$
④ Final addition (Beware of impulsive $18 + 5$ slip!):
 $18 + (\quad) =$

11. Evaluate: $4.8 \times 2.5 + 5.2 \times 2.5$ using the distributive property.

DISTRIBUTIVE PROPERTY STEPS

=

SCRATCH COLUMN

- $4.8 + 5.2 =$
 $10 \times 2.5 =$

Section 4: SUIS Integrated Track Challenge Problems (15 pts each, 30 pts total)

Score: _____

12. [Divisibility & Modulo Dual-Variable Problem] The six-digit number $\overline{91X93Y}$ is divisible by both 8 and 9 (i.e. divisible by 72). Find all possible values of X and Y .

Follow the standard three-step pipeline. DO NOT stop after solving for Y !

Step 1 (Divisibility by 8 via last 3 digits):

For a number to be divisible by 8, its last 3 digits $\overline{93Y}$ must be divisible by 8.

$93Y = 930 + Y = 8 \times 116 + (2 + Y)$.

Since $2 + Y$ must be a multiple of 8 and $Y \in \{0, 1, \dots, 9\}$, we have $Y =$ _____.

Step 2 (Divisibility by 9 via sum of digits):

Now the number is $\overline{91X93Y}$.

The sum of digits $S = 9 + 1 + X + 9 + 3 + (\quad) =$ _____.

For S to be divisible by 9, $X =$ _____.

Step 3 (Final Answer):

Answer: $X =$ _____, $Y =$ _____, and the six-digit number is _____.

13. [Paper Folding & Exponential Growth Model] A sheet of paper has thickness 0.1 mm. Suppose it can be folded in half repeatedly.

(a) Folded 1 time, thickness is $0.1 \times 2 = 0.2$ mm. Folded 3 times, number of layers is $2^3 =$ _____, thickness is _____ mm.

(b) If folded 10 times, what will the thickness be in millimeters and in meters? (Given $2^{10} = 1024$)

(a) Layers after 3 folds: $2^3 =$ _____; Thickness = $0.1 \times 2^3 =$ _____ mm.

(b) Thickness after 10 folds:

$0.1 \text{ mm} \times 2^{10} = 0.1 \times 1024 =$ _____ mm = _____ m.



[Sophia Set 1] Answer Key & Scoring Guide (For Review)

Section 1: Conceptual Precision (25 pts)

1. (8 pts) Square numbers: $1 = 1^2$, $4 = 2^2$, $9 = 3^2$, $16 = 4^2$, $25 = 5^2$, $36 = 6^2$, $49 = 7^2$, $64 = 8^2$, $81 = 9^2$.

Note: 6, 12, 18, 24, 48, 72 must NOT be circled!

2. (5 pts) False (2 pts). Reason: A square number must be the product of two identical integer factors ($a \times a$). For $6 = 2 \times 3$, factors 2 and 3 are different (3 pts).

3. (5 pts) (a) 6 cm (2 pts), because $6 \times 6 = 36$ (1 pt). [DO NOT write $36 \div 2 = 18$]. (b) Perimeter = $4 \times 6 = 24$ cm (2 pts).

4. (5 pts) (a) Area = $8 \times 2 = 16 \text{ cm}^2$ (2 pts). (b) Yes (1 pt), side length = 4 cm, because $4^2 = 16$ (2 pts).

5. (5 pts) $180 = 2^2 \times 3^2 \times 5^1$ (2 pts). To make all exponents even, we need one more 5, so $k = 5$ (3 pts).

Section 2: Equivalent Fractions (25 pts)

6. (5 pts) (a) $\frac{3 \times 5}{4 \times 5} = \frac{15}{20}$; (b) $\frac{2 \times 4}{5 \times 4} = \frac{8}{20}$; (c) $\frac{15}{20} + \frac{8}{20} = \frac{23}{20} = 1\frac{3}{20}$.

7. (5 pts) Error: Only the denominator was multiplied; the numerator was left unchanged (2 pts). Correct: $\frac{10}{12} - \frac{3}{12} = \frac{7}{12}$ (3 pts).

8. (10 pts) (a) LCM[8, 6] = 24; Expression = $\frac{15}{24} + \frac{4}{24} = \frac{19}{24}$ (5 pts). (b) Expression = $\frac{7}{3} - \frac{7}{4} = \frac{28}{12} - \frac{21}{12} = \frac{7}{12}$ (5 pts).

9. (5 pts) $1 - \frac{11}{13} = \frac{2}{13}$, $1 - \frac{15}{17} = \frac{2}{17}$. Since $\frac{2}{13} > \frac{2}{17}$, the distance to 1 is larger, so $\frac{11}{13} < \frac{15}{17}$.

Section 3: Two-Column Scratch Work (20 pts)

10. (10 pts) Expression = $18 + 5 \times 6 \div 3 = 18 + 30 \div 3 = 18 + 10 = 28$. Full scratch work earns full credit.

11. (10 pts) Expression = $(4.8 + 5.2) \times 2.5 = 10 \times 2.5 = 25$.

Section 4: Extension Problems (30 pts)

12. (15 pts) Step 1: $Y = 6$ (5 pts); Step 2: Sum of digits $S = 9 + 1 + X + 9 + 3 + 6 = 28 + X$. For S to be divisible by 9, $28 + X = 36 \implies X = 8$ (7 pts); Step 3: $X = 8$, $Y = 6$, number is 918936 (3 pts).

13. (15 pts) (a) 8 layers (2 pts), 0.8 mm (3 pts). (b) Thickness = $0.1 \times 1024 = 102.4 \text{ mm} = 0.1024 \text{ m}$ (10 pts).